Orbit determination and control system

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1 Introduction

Orbit determination and control are well-established concepts in traditional space missions [1–4]. These techniques cover a wide range of applications, from ground based to onboard orbit determination, and from single spacecraft to formation flying control. Early CubeSat missions did not include these capabilities onboard, because of the scarce resources and the simplicity of the missions. Orbit determination was provided by propagating ephemerides with commercial software.

The growth of CubeSat missions in terms of complexity and capabilities has required the development of dedicated orbit determination and control systems (ODCS). Nowadays, in fact, many missions are using or planning to use these devices in order to reach their mission's goals or simply to test new products and equipment on orbit. For example, orbit determination and control might be required for performing certain tasks at specific points on the orbit or to make changes to the trajectory. These are key factors in the development of CubeSat constellations, which have been proposed and launched for several purposes, such as Earth observation, atmospheric measurements, surveillance, and disaster management [5–7]. Achieving and maintaining specific orbits requires the capability of performing orbital maneuvers and navigating autonomously, even without GPS or Earth's magnetic field, as in the case of deep space missions. CubeSat missions outside of Earth orbit allow mission operators to undertake science measurements in several areas at an affordable cost and, at the same time offer more redundancy and increased launch opportunities compared to traditional satellites. At the end of the 2010s, CubeSat interplanetary missions have been successfully launched or are in preparation for the first time in history. In 2018, the MarCO mission was the first CubeSat interplanetary mission to be launched. It consisted of two CubeSats that completed a Mars fly-by, serving as a radio link provider for the InSight Mars lander. The radio link was also used to determine the spacecraft positioning. Another relevant initiative is the maiden launch of the Artemis 1 mission, currently planned for 2021, that will carry 13 6U CubeSats to lunar orbit, to an asteroid encounter and other destinations, with different solutions for orbit control and determination. For further details on interplanetary missions, the reader is referred to the chapter of this book dedicated to this topic.

CubeSat Handbook. https://doi.org/10.1016/B978-0-12-817884-3.00013-8

This chapter is organized as follows: Section 2 describes the main orbit determination techniques, along with the main algorithms; Section 3 describes the main orbit control technologies, the maneuvers, and the implementation of control commands.

2 Orbit determination

Determining the orbit of a spacecraft involves reconstructing its trajectory from a set of measurements. A distinction can be made between what is usually called *preliminary orbit determination* and *orbit estimation* [2]. The former consists of processes associated with defining the full set of orbital parameters from six observations. The latter is the reconstruction of the orbit from a large set of measurements, and from the use of numerical techniques. For CubeSats, preliminary orbit determination is useful when TLEs (two-line elements) are not yet available (e.g., at the release from the launcher). Orbit estimation is instead crucial for all the rest of the mission phases, particularly for cases in which orbital knowledge is of interest to the mission itself.

The methods of orbit determination can be divided between ground-based and onboard technologies. In general, the spacecraft is tracked from a network of ground stations that provide an estimate of its orbit. However, some missions may require an onboard orbit determination system for several reasons (e.g., visibility of the spacecraft from the Earth, requested accuracy, short-time scale of the orbital motion, etc.). Orbit determination can be performed in several ways, depending on what measurements are available.

2.1 Ground-based technologies

The simplest and most employed method of orbit determination for CubeSat missions has been to propagate ephemeris with numerical orbital models. The most widely used common set of ephemerides are the TLEs, which are issued by the North American Aerospace Defense Command (NORAD) for a large number of satellites and objects orbiting the Earth [8]. TLEs are compatible with many orbital propagation models, such as SGP-4 and SDP-4, and include terms that account for the atmospheric drag effects. The position of a 3U CubeSat reconstructed with NORAD TLEs is reported to be accurate to 1 km and grows approximately 1–2 km per day [9]. More accurate TLEs can be generated by radio-frequency ranging as in the case of the Planet CubeSats [10].

The propagation of the TLEs can happen on board or on the ground. Ground-based propagation is used either when orbit information is not needed on board or when it is sufficient to transmit the propagated orbit once the spacecraft is passing over an available ground station. Onboard propagation requires a computer capable of running an orbital model and a radio to receive updated ephemeris.

Ground tracking methods, including optical tracking through LED [11] and laser illuminated retroreflectors [12], have also been recently proposed for CubeSats.

2.2 Onboard technologies

Onboard orbit determination using Global Navigation Satellite Systems (GNSS) has recently found application in CubeSat missions [13, 14]. Commercial Off-The-Shelf (COTS) receivers for CubeSats are currently available on the market, including dual-frequency and multiconstellation receivers. In some cases, they need to be modified in order to remove the speed and altitude limitations imposed by the manufacturers that prevent the use in space missions. If available from the onboard receiver, GNSS raw measurements can be numerically processed on board to reconstruct the orbit with a high accuracy. The major sources of error in the GNSS signals are clock bias, ionospheric, tropospheric, and multipath errors. Some of these errors can be canceled by processing the measurements with proper algorithms or by using differential GNSS [15].

Another onboard orbit determination technology is optical navigation. It employs cameras to position the spacecraft with respect to some known objects. In case the position is calculated in a local reference frame, a second instrument (e.g., a star tracker) is needed to position the spacecraft in an inertial frame. The advantage of this technology is that it allows autonomous navigation in deep space missions with a sufficient level of accuracy. Furthermore, it can be used in rendezvous or flybys in proximity of the targets, as in the case of the DustCube mission. DustCube uses two infrared cameras to calculate the line of sight from the spacecraft to the two asteroids that are targeted for study; the position of the spacecraft is then calculated at the geometrical interception of the two beacons [16]. The Cislunar explorers are two other CubeSats that plan to use optical navigation. They are equipped with commercial cameras to reconstruct their position from images of the Earth, the Moon, and the Sun. The sizes of these objects and their relative displacements are used to calculate the distance of the spacecraft from these known objects [17].

Sometimes calculation of the orbital position in an absolute reference frame is not needed, but calculation of the relative motion with respect to some other body (another spacecraft or a celestial body) is required. This is the case, for example, of two satellites in a leader-follower configuration, where one is tracked with high precision (from ground or with a GNSS receiver) and the orbit of the latter is reconstructed with relative measurements from the former. Some recent CubeSat missions have proposed the use of intersatellite links: the RANGE mission will use a compact on board laser ranging system to implement formation flying [18]; the AAReST mission will use an active lidar sensor for relative navigation between CubeSats that compose a reconfigurable space telescope [19].

2.3 Algorithms of orbit determination

The first algorithms of orbit determination date back to the early years of the 19th century with the method described by Gauss for preliminary orbit determination. To obtain the six orbital parameters, six independent observations are needed, for example, two position vectors or three direction vectors [2]. These algorithms are based on geometrical rules and are quite sensitive to measurement noise and modeling uncertainties that are likely to affect the observations. Nevertheless, they can be used for preliminary orbit determination.

Orbit estimation usually relies on a large amount of data. In this case numerical algorithms are employed to cancel the effects of noise and uncertainties. A number of batch and recursive algorithms are used toward this end [2]. Batch algorithms are based on least-squares estimation and are usually employed at ground for solving the preliminary orbit determination problem. Recursive algorithms are more likely to be employed on board, due to the limited computational burden and the production of an online solution. Kalman filtering is the common framework for recursive orbit determination.

For both batch and recursive algorithms, the problem of orbit estimation consists in determining the best estimation for the state vector x

$$\vec{x}\,(\vec{x}) = \left(\vec{r}\,(t)\;\;\vec{v}\,(t)\;\;\vec{p}\,\right)^T \tag{1}$$

where $\vec{r}(t)$ and $\vec{v}(t)$ are the satellite's position and velocity, respectively, and $\vec{p}$ is a vector of parameters that possibly affect the model (e.g., the ballistic coefficient or the GNSS receiver bias). The time evolution of $\vec{x}$ follows a generic nonlinear model

$$\dot{x}(t) = f(t,x) \tag{2}$$

The available observations $\vec{z}$ are described by another nonlinear model corrupted by the noise $\vec{\nu}$ due to measurement errors

$$z(t) = h(t,x) + \nu(t) \tag{3}$$

2.3.1 Batch estimation

The most widely used common batch estimation method is the weighted least-squares algorithm. It consists in finding the value of the state at the reference epoch t_0, $x_0 = \vec{x}(t_0)$, which minimizes the quadratic cost function

$$J(x_0) = (z(t_0) - h(t_0,x))^T (z(t_0) - h(t_0,x)) \tag{4}$$

over a set of given measurements $\vec{z}$. Defining H as the Jacobian matrix of $\vec{h}$ on a reference trajectory

$$H = \left.\frac{\partial h(t_0, \vec{x})}{\partial x_0}\right|_{x_0 = x_0^{ref}} \tag{5}$$

the cost function can be rewritten in terms of the differences $\Delta x_0 = x_0 - x_0^{ref}$ and $\Delta z_0 = z(t_0) - h(x_0^{ref})$ as

$$J(\Delta x_0) = (\Delta z_0 - H\Delta x_0)^T (\Delta z_0 - H\Delta x_0) \tag{6}$$

The minimum is found by the condition that $\frac{\partial J(\Delta x_0)}{\partial \Delta x_0} = 0$ and the least-squares solution is

$$\Delta x_0^{lsq} = (H^T H)^{-1} (H^T \Delta z_0) \tag{7}$$

Since not all measurements have the same accuracy, they can be weighted by normalizing the residuals in accordance with the respective noise variances $\sigma_1, \ldots, \sigma_n$. Defining the matrix $S = diag(\sigma_1^{-2}, \ldots, \sigma_n^{-2})$, the weighted least-squares solution can be defined once again as

$$\Delta x_0^{wlsq} = (H^T S H)^{-1} (H^T S \Delta z_0) \tag{8}$$

2.3.2 Filtering

Real-time measurement processing requires the use of a sequential algorithm. Among the available solutions, the Kalman filter (KF) family of algorithms provides consolidated real-time, recursive state estimation from noisy measurements [2, 20]. The KF algorithm is composed of two phases, prediction and correction. Within the prediction phase, the current estimate $\hat{x}$ and the estimation error covariance P are propagated forward in time using the dynamic model of the system in Eq. (2). Within the correction phase, the propagated estimate and covariance are corrected with a term that weighs the difference between the noisy measurements from the sensors and the measurements that the estimated state would produce.

The original formulation of the KF is suitable for linear systems. When it comes to nonlinear applications, such as orbit estimation, more advanced algorithms are required. The extended Kalman filter (EKF) was developed during the Apollo program exactly for this purpose. Its main feature is the linearization of the dynamical model around the actual conditions at each time step. This allows the mission designers to approximate and propagate the mean and the covariance of the state variables as in the KF equations. Other nonlinear algorithms have been proposed that improve EKF's performance in terms of robustness and accuracy. The unscented Kalman filter (UKF) uses a finite number of state-space points to propagate the nonlinear model, avoiding the EKF linearization. The particle filter (PF) is a probability-based estimator that recursively implements Monte-Carlo-based statistical signal processing. One of the drawbacks of these two algorithms is that they generally require more computational power than the EKF. The improvements in CubeSat computers capacities in this sense has made possible their use onboard.

The equations associated with the above-mentioned algorithms will not be reported here for the sake of conciseness as they can be found in Refs. [2, 20]. The rest of this section is dedicated to showing some of the issues related to orbit reconstruction with filters. A first important point is related to the kind of model provided to the filter: the main types are pure kinematic, pure dynamic, and reduced dynamic. The first does not require a representation of forces, but is very sensitive to measurements; the second is difficult to implement, especially in low orbits, because disturbances are hard to model (e.g., atmospheric drag); the third is perhaps the most practical, because disturbances can be modeled as empirical accelerations through the process noise covariance matrix Q of the filter [21].

The Q matrix is one of the most important tuning parameters of a KF. It represents the level of uncertainty in the model. Even though CubeSats have standard dimensions and shape, modeling the effects of perturbations such as atmospheric drag, third bodies, solar radiation pressure, and so on is particularly difficult and not so cost-effective in terms of computational burden. It is easier to consider a model of motion represented by the state $x(t) = \begin{pmatrix} r(t) & v(t) \end{pmatrix}^T$ and to inject noise in the model through the Q matrix to account for these effects. Q is defined as

$$Q = \int_0^{T_s} \Phi(\eta)\Omega\Phi(\eta)^T d\eta \tag{9}$$

where Φ is the state transition matrix of the model, T_s is the sampling time of the measurements, and Ω is the matrix whose structure reflects how the noise enters in the dynamics. In the case of a simple position-velocity state vector, where the noise enters the model as a disturbance acceleration, Ω is defined as

$$\Omega = q_0 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{10}$$

where q_0 is a numerical value representing the level of noise. Finally, the structure of Q in this case is

$$Q = q_0 \begin{pmatrix} T_s^3/3 & T_s^2/2 \\ T_s^2/2 & T_s \end{pmatrix} \tag{11}$$

By adjusting the value of q_0 the effects of unmodeled disturbances can be accounted for in the algorithm.

An example of parameter estimation with the KF is the evaluation of the GNSS constant clock bias b. To do that it suffices to add to the filter model in Eq. (2) a null differential equation:

$$\dot{b} = 0 \tag{12}$$

and to adjust the Q matrix structure in accordance.

3 Orbit control

Orbit control generally involves maneuvers that alter a spacecraft's trajectory and involve some sort of propulsion: orbit parameters changes (e.g., inclination), altitude raising, deorbiting, formation control, station keeping, etc. Depending on the goals of the mission, different technologies might be more or less adequate.

3.1 Orbit control technologies

Orbit control has not been employed in many CubeSat missions so far, due to the limitation of resources and the absence of specific mission requirements. Nevertheless, several different technologies have been proposed for use [22]. Orbit control technologies can be divided into active and semiactive: the former directly spend energy in order to produce a momentum change in the spacecraft; the latter employs some sort of energy to indirectly change the orbit.

In most spacecraft, orbit control is provided by some sort of propulsion (e.g., chemical, electrical, etc.). However, propulsion solutions for CubeSats must comply with several limitations and restrictions at the same time, as, for example, the prohibition of pyrotechnics on board, the limitation on the amount of chemical propellant, and the safety procedures relating to pressurized tanks. For these reasons, only cold-gas thrusters and electric propulsion have been employed in CubeSat missions to date.

Alternative, propellantless methods to traditional propulsion systems have been proposed and launched, such as solar sails [23] and differential drag [10]. Solar sails collect photons emitted by the Sun in order to produce a propulsive force. The sail should be large enough to guarantee a sufficient momentum change. Differential drag is a method of controlling a spacecraft ballistic coefficient in order to produce a suitable amount of drag force. The ballistic coefficient can be changed with an attitude maneuver of the spacecraft so that the aerodynamic cross-section of the spacecraft changes. This can be used to space two CubeSats along the same orbit. Another way of obtaining this effect is to release the CubeSats at different points of the same orbit, if the dispenser has the possibility to do so [24].

3.2 Maneuvers

Orbital maneuvers can be divided into in-plane and out-of-plane maneuvers. In-plane maneuvers are produced by a combination of forces tangential and normal to the trajectory. They result in a change of the semimajor axis a, the eccentricity e, and the argument of perigee ω of the orbit. In the case of CubeSats, the most common in-plane maneuvers are, for example, the altitude raising and reentry. These maneuvers are directly related to variations of the semimajor axis a and therefore to tasks such as deorbiting and mission life extension.

Changes in a are produced by altering the speed of the spacecraft. From the *vis viva* equation, it is known that the orbital speed v in an elliptical orbit is

$$v = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)} \tag{13}$$

Differentiating the two sides, a change in velocity Δv is related to a variation Δa:

$$\Delta v = \frac{\mu}{2va^2}\Delta a \tag{14}$$

Since in general orbits do not have constant speed, the minimum required Δv to obtain a certain Δa is when v is maximum, that is, at the perigee.

When the Δv is impulsively applied to the spacecraft in the direction of the speed, that is, after the activation of a thruster, the semimajor axis changes instantaneously and the spacecraft is transferred to another orbit. The simplest example of orbital transfer is that between two circular orbits (a higher and a lower one), called the Hohmann transfer. In its minimal configuration, shown in Fig. 1, it is composed of two impulsive maneuvers Δv_1 and Δv_2: the first takes the spacecraft on an elliptical orbit with perigee at point 1; the second circularizes the orbit at the altitude of the apogee of the transfer orbit. Other similar methods are the bi-parabolic, bi-elliptical, and multiple impulse Hohmann transfers. The performance of these methods in terms of required Δv depend on the ratio between the final radius and starting radius of the orbits, with the Hohmann transfer being the best solution for $r_2/r_1 \leq 11$.

Out-of-plane maneuvers are produced by forces perpendicular to the orbital plane. The effect is a variation of the inclination i of the orbital plane. Changing only the orbital plane requires that the orbital velocity vector is rotated by an angle Δi maintaining its absolute value. Observing Fig. 2, the Δv required to perform the maneuver is calculated with simple trigonometric rules as

$$\Delta v = 2v \sin\frac{\Delta i}{2} \tag{15}$$

3.3 Commands implementation

The velocity variations calculated in the previous sections can be seen as *commands* to the propulsion system. A few considerations have to be taken into account in order to implement these commands. From the mission planning point of view, for all those

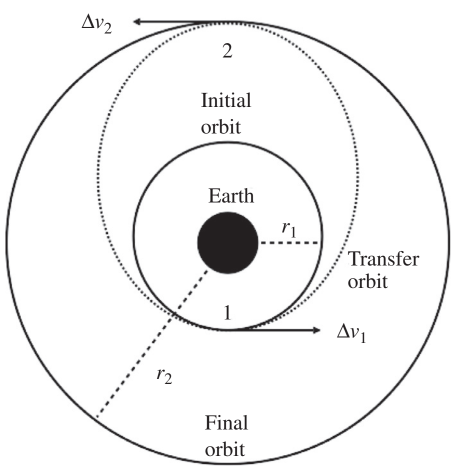

Fig. 1 The Hohmann transfer.

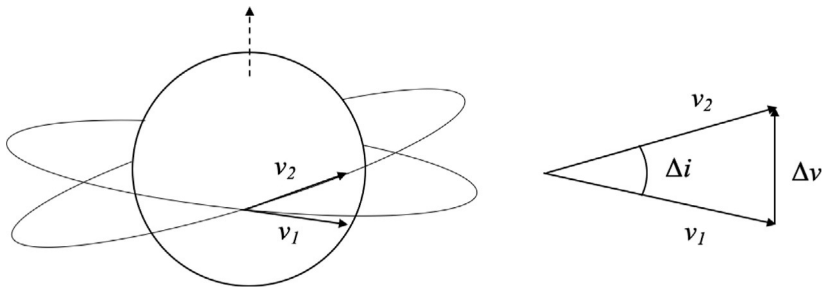

Fig. 2 Orbital plane change maneuver.

spacecraft using some sort of propellant, it must be considered that its onboard availability is limited. Therefore, it is fundamental to establish in advance the total amount of the propellant required for the entire mission. This quantity is defined as the Δv budget. First of all, Δv is calculated from the knowledge of the actual orbital state of the spacecraft, which is precise only up to a certain level of accuracy. Another important fact is that impulsive Δvs cannot be implemented by any real propulsive system, which is characterized by finite time responses and delays. Furthermore, each propulsive system might present errors relating to operating conditions, degradation, unmodeled behaviors, and precision.

A Δv command is implemented in a series of firings or as a continue force in the case of solar sails. Neglecting mass variation, the second law of dynamics can be written as

$$f\Delta t = m\Delta v \tag{16}$$

where f is the total force from the propulsion system, Δt is the total thrusting time, and m is the mass of the spacecraft. Splitting the force in multiple impulses, as in the case of the multi-impulse Hohmann transfer, one obtains

$$\sum_i f_i \Delta t_i = m \sum_i \Delta v_i \tag{17}$$

The duration of the thrust varies with the propulsion system: electric propulsion has very short firings repeated for very long time, while other kinds of propulsion may have less firings. After a single firing, the spacecraft status might be checked to correct the errors in the successive maneuver. This is usually performed on the ground by the mission operations team. Of course, this open-loop strategy might not be adequate in the case of very strict requirements related to the precision and timing.

A different, onboard-based approach is to build a control loop around the actuator, which can be also characterized by a transfer function between the command and the implemented force. Here, onboard accelerometers would be employed as a feedback measurement to calculate the error between the command and the actual values provided by the actuator [4]. Decisions regarding which of these orbit determination and control methods to use should be carefully considered and driven by the mission requirements.

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